ASSIGNMENT-4 SOLUTIONS: Symmetric Cryptography

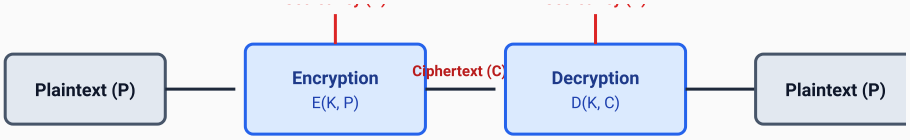
COURSE	SUBJECT CODE	SUBJECT TITLE	TOPIC
BCA	BCA309T-AS04	Cryptography & Network Security	Introduction to Symmetric Cryptography

Q1. Principles of Symmetric Cryptography & Kerckhoffs's Principle [10 Marks] LONG ANSWER

Basic Principles: Symmetric cryptography (secret-key cryptography) uses a single secret key (K) for both encryption and decryption. Both sender and receiver must share this key securely prior to transmission.

- **Plaintext (P):** The original, readable message (e.g., `"TRANSFER 1000"`).
- **Secret Key (K):** A secret value known only to authorized parties that configures the algorithm.
- **Encryption Algorithm (E):** Mathematical operations applied to plaintext using the key: $C = E(K, P)$.
- **Ciphertext (C):** Scrambled, unreadable output transmitted over insecure networks.
- **Decryption Algorithm (D):** Reverses encryption using the same key to recover text: $P = D(K, C)$.

Kerckhoffs's Principle: Stated by Auguste Kerckhoffs in 1883, it dictates that *a cryptosystem should remain secure even if everything about the system (including algorithm design) is public knowledge, except for the secret key.* Security must rely solely on key secrecy, not "security through obscurity."



Q2. Stream Ciphers vs Block Ciphers, Confusion/Diffusion & Block Cipher Modes [10 Marks] LONG ANSWER

Stream Ciphers vs. Block Ciphers: Stream ciphers encrypt plaintext continuously bit-by-bit or byte-by-byte using a pseudorandom keystream (e.g., RC4). Block ciphers process fixed-size blocks of plaintext simultaneously (e.g., 64-bit in DES, 128-bit in AES).

Confusion and Diffusion (Claude Shannon):

- **Confusion:** Complexifies the relationship between ciphertext and key. Implemented via *substitution* (e.g., S-Boxes) so key bits cannot be deduced easily.
- **Diffusion:** Spreads statistical structure of plaintext across ciphertext. Implemented via *transposition/permutation* (e.g., P-Boxes, ShiftRows) so changing 1 plaintext bit alters ~50% of ciphertext bits.

Main Block Cipher Modes of Operation:

Mode	Working Mechanism	Strengths	Weaknesses
ECB (Electronic Codebook)	Each block encrypted independently using same key: $C_i = E(K, P_i)$.	Simple; allows parallel processing; error confined to 1 block.	Insecure for patterned data; identical plaintext blocks yield identical ciphertext.
CBC (Cipher Block Chaining)	Plaintext XORed with previous ciphertext block: $C_i = E(K, P_i \oplus C_{i-1})$. Uses IV.	Hides plaintext patterns completely; widely supported standard.	Sequential encryption (cannot parallelize); error propagates to next block.
CFB (Cipher Feedback)	Encrypts previous ciphertext to form keystream stream cipher: $C_i = P_i \oplus E(K, C_{i-1})$.	Converts block cipher to stream cipher; processes small units (bytes).	Sequential encryption; bit error in ciphertext affects current & next block.
OFB (Output Feedback)	Encrypts internal feedback buffer continuously to create keystream: $S_i = E(K, S_{i-1})$.	No error propagation; keystream pre-computable offline.	Vulnerable to bit-flip attacks if message authentication is absent.
CTR (Counter)	Encrypts incrementing counter values to create keystream: $C_i = P_i \oplus E(K, \text{Counter}_i)$.	High speed; fully parallelizable for both encryption & decryption.	Counter reuse completely destroys security (never reuse IV/Counter).

Q3. Historical Development of Symmetric Encryption Algorithms [10 Marks] LONG ANSWER

Historical Evolution Timeline:

1. **Early Classical Era:** Character-based systems like the *Caesar Cipher* (monoalphabetic shift) provided primitive secrecy. The 16th-century *Vigenère Cipher* introduced polyalphabetic keywords to resist simple frequency analysis.
2. **Mechanical Era & WWII:** Rotor machines like Germany's *Enigma* automated polyalphabetic substitution. Cryptanalysts like Alan Turing developed mechanical computing engines (the *Bombe*) to break rotor settings.
3. **Shannon's Information Theory (1949):** Claude Shannon published a mathematical foundation establishing *confusion and diffusion* as core design requirements for secure ciphers.
4. **Feistel Cipher & DES (1970s):** Horst Feistel developed a symmetric structure splitting data into halves. IBM built Lucifer, which NIST standardized in 1977 as **DES** (56-bit key, 64-bit block).
5. **Triple DES & Alternative Block Ciphers:** When 56-bit DES keys became vulnerable to brute-force, **3DES** (using 2 or 3 keys) extended life. Algorithms like **IDEA** (128-bit key), **Blowfish** (variable key), and **RC4** (stream cipher) emerged.
6. **The AES Open Competition (1997–2001):** NIST hosted a public multi-year contest to replace DES. The **Rijndael cipher** (by Vincent Rijmen & Joan Daemen) won due to top hardware/software speed, non-Feistel SPN structure, and strong security, becoming **AES** in 2001.

Q4. Classical Substitution Ciphers & Frequency Analysis [10 Marks] LONG ANSWER

1. Caesar Cipher: Shifts every letter by a fixed key k: $C = (P + k) \pmod{26}$.
• *Weakness:* Only 25 possible keys; trivial brute-force search.

2. Monoalphabetic Substitution Cipher: Maps each letter of the alphabet to a unique random letter permutation.
• *Strengths:* $26! \approx 4 \times 10^{26}$ key space prevents brute force.
• *Weaknesses:* Preserves letter frequency distribution of natural language completely.

3. Playfair Cipher: Encrypts letter pairs (digrams) using a 5×5 matrix built from a key word.
• *Working:* If letters in same row, shift right; if same column, shift down; if rectangle, swap corners.
• *Strengths:* Flattens single-letter frequencies (676 digrams instead of 26 single letters).

4. Vigenère Polyalphabetic Cipher: Uses a repeating keyword to apply different Caesar shifts per position: $C_i = (P_i + K_i) \pmod{26}$.
• *Strengths:* Masks single-letter frequencies because 'E' encrypts to different ciphertext letters depending on keyword position.

Why Frequency Analysis Works / Fails: Frequency analysis measures letter occurrences (e.g., 'E', 'T', 'A' are most common in English). It easily breaks Caesar and Monoalphabetic ciphers because letter probabilities remain unchanged. It is less effective against Playfair and Vigenère because polyalphabetic substitution masks letter frequencies across multiple shifting keys.

Q5. Transposition Techniques, Product Ciphers & One-Time Pad [8 Marks] LONG ANSWER

Transposition Ciphers: Rearrange letter positions without altering identities.
• *Rail Fence:* Writes text diagonally across N rails and reads row-wise.
• *Columnar Transposition:* Writes text in a grid under a key phrase, reading columns based on alphabetical key order.

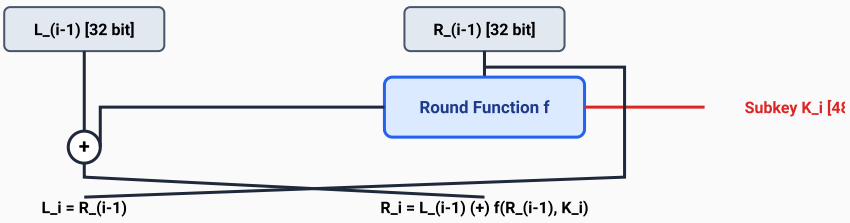
Product Ciphers: Combine substitution (confusion) and transposition (diffusion) sequentially. Combining two weak ciphers creates a cipher far stronger than either alone. This idea forms the foundation of modern block ciphers like DES and AES (Substitution-Permutation Networks).

One-Time Pad (OTP): Encrypts text with a truly random key of equal length: $C_i = P_i \oplus K_i$.
• *Unbreakable:* Mathematically perfect secrecy (proven by Shannon) because ciphertext yields no information.
• *Impractical:* Key must be as long as message, used once only, and shared securely beforehand.

Q6. Data Encryption Standard (DES) Structure & Round Function [12 Marks] LONG ANSWER

DES Parameters: 64-bit block size, 56-bit effective key size (8 bits parity), 16 Feistel rounds.

Round Operations: $L_i = R_{i-1}$, $R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$. The round function f expands 32-bit R_{i-1} to 48 bits, XORs with 48-bit subkey K_i, passes through 8 S-Boxes (48$\to$32 bits), and applies 32-bit P-Box permutation.



Q7. Strengths & Weaknesses of DES, 3DES & Meet-in-the-Middle Attack [10 Marks] LONG ANSWER

Strengths & Weaknesses of DES: DES exhibits a strong *avalanche effect* (flipping 1 input bit alters over 50% of output bits) and strong S-Box design against differential cryptanalysis. However, its **56-bit key size** is critically short ($2^{56} \approx 7.2 \times 10^{16}$ keys), making it easily crackable via specialized hardware brute-force in hours.

Double DES & Meet-in-the-Middle (MITM) Attack: Double DES encrypts twice: $C = E(K_2, E(K_1, P))$. Though key length is 112 bits, it provides only 57-bit security due to the MITM attack. An attacker computes intermediate values $X = E(K_1, P)$ forward for all 2^{56} keys and decrypts $X' = D(K_2, C)$ backward for all 2^{56} keys. Matching table entries reveals (K_1, K_2) in 2^{57} operations rather than 2^{112}.

Triple DES (3DES): Overcomes MITM using 3 operations: $C = E(K_3, D(K_2, E(K_1, P)))$. 3-key 3DES provides 168-bit key size (112-bit effective security) and remains secure, though computationally slow.

Q8. Advanced Encryption Standard (AES) Architecture & Comparison with DES [10 Marks] LONG ANSWER

AES Key Parameters: Fixed 128-bit block size. Supports key lengths of 128-bit (10 rounds), 192-bit (12 rounds), and 256-bit (14 rounds).

Four Round Transformations:

1. **SubBytes:** Non-linear byte substitution using S-Box over $GF(2^8)$ (provides confusion).
2. **ShiftRows:** Cyclically shifts rows of 4×4 State matrix (provides diffusion).
3. **MixColumns:** Matrix multiplication over Galois Field transforming each column (diffusion).
4. **AddRoundKey:** XORs State matrix with 128-bit round subkey derived from key expansion.

Comparison Matrix: AES vs. DES:

Feature	Data Encryption Standard (DES)	Advanced Encryption Standard (AES)
Architecture	Feistel Cipher Network (half-block processing).	Substitution-Permutation Network (SPN, full block).
Block Size	64 bits.	128 bits.
Key Lengths	56 bits (vulnerable to brute force).	128, 192, 256 bits (highly secure).
Hardware / Software Speed	Slow in software; optimized for 1970s hardware.	Exceptionally fast in both hardware and software.
Cryptanalytic Resistance	Vulnerable to linear attacks and brute force.	Resistant to all known linear, differential & brute-force attacks.